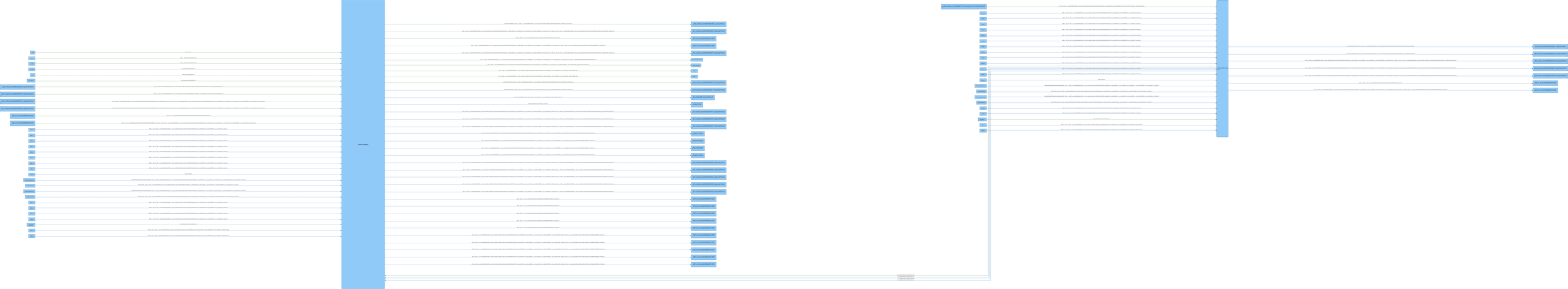
transfer function